

Samiul Hasan

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ABOUT ME

Electrical and Electronic Engineering graduate with practical experience in electrical project planning, material estimation, technical documentation, and site inspections. My interests include sustainable energy systems, advanced power electronics, and optical sensing technologies, with a strong interest in applying engineering knowledge to practical research and development.

EDUCATION AND TRAINING

2022/01/20 – 2025/12/31 Dhaka, Bangladesh
BACHELOR OF SCIENCE (B.SC.) Independent University, Bangladesh (IUB)

Website https://iub.ac.bd/ | **Field of study** Electrical and Electronic Engineering (EEE) | **Final grade** 3.32 | **Level in EQF** EQF level 6 |

Number of credits 147 | **Thesis** Numerical Investigation and Performance Enhancement of a PV-Assisted Solar Thermal Dehydrator Using CFD

WORK EXPERIENCE

INTERN ENGINEER – PACIFIC POWER TECH – 2026/08/01 – Current – DHAKA, BANGLADESH

1. Prepared electrical Single Line Diagrams (SLDs) and distribution layouts using AutoCAD Electrical. Conducted substation site inspections covering transformers, HT/LT switchgear, circuit breakers, busbars, and grounding systems.
2. Prepared BOQs, material takeoffs, cost estimates, and procurement specifications for power distribution projects.
3. Designed a 2,232 sq. ft. rooftop solar PV system using PVsyst and supported post-fire electrical inspection and insulation resistance testing.

RESEARCH ASSSITANT – INDEPENDENT UNIVERSITY,BANGLADESH – 2025/08/05 – Current – DHAKA , BANGLADESH

1. Conducted CFD and multiphysics thermal simulations, data analysis, and performance evaluation for solar energy applications.
2. Contributed to research in optical sensing, air-quality monitoring, machine learning, and power electronics.
3. Assisted faculty supervisors with research documentation, manuscript preparation, technical paper checking, data analysis, and literature review. Contributed to IEEE conference publications and peer-reviewed manuscripts, developing skills in technical writing and research methodology.

UNDERGRADUATE TEACHING ASSISTANT – INDEPENDENT UNIVERSITY,BANGLADESH – 2025/02/20 – 2025/05/28 – DHAKA, BANGLADESH

1. Supervised 30+ undergraduate students in Power Electronics and Circuit Simulation laboratories, supporting hands-on learning and technical problem-solving. Guided students in circuit design, numerical analysis, simulation, and hardware troubleshooting using MATLAB/Simulink, PSIM, and Proteus.
2. Assessed laboratory work and developed simulation exercises and problem-solving sessions, strengthening students' analytical and practical engineering skills.

SKILLS

Simulation & Design Tools:

Cadence Virtuoso (GPDK90nm) | Python | MATLAB | AutoCAD Electrical (2D) | LaTex | Comsol Multiphysics | PVsyst

Language:

English (Elementary) | Bengali

HONOURS AND AWARDS

Efficiency for Access Design Challenge UK – IKEA Foundation Grant (2024 – 2025)

Awarded by *Energy Saving Trust & IKEA Foundation.* Selected and awarded grant funding of BDT 35,000 to develop an off-grid solar thermal dehydrator for rural agricultural preservation and clean energy access.

Champion – Line Following Robot Competition (August 2024)

IUB EEE Innovation Fest 2024, Independent University, Bangladesh.

PROJECTS

2025/02/06 – 2025/06/15
Design and Implementation of a Two-Wheeled Self-Balancing Robot

Designed and developed a two-wheeled self-balancing robot using PID control and motor driver circuits, implementing real-time feedback control for stability, balance correction, and motion control under dynamic conditions.

2025/04/10 – 2025/06/28
Simulation-Based Design and Performance Analysis of a 310V/19V 500W Dual-Stage Buck Converter

Designed and simulated a 310 V/19 V, 500 W dual-stage MOSFET-based DC–DC buck converter with symmetrical duty cycles operating at 20 kHz.
Analyzed converter performance under varying resistive loads to evaluate efficiency, voltage regulation, and system stability through simulation.

RESEARCH EXPERIENCE & PUBLICATIONS

Peer-Reviewed Journal Articles (Under Peer Review)

1. **Samiul Hasan**, Fardous Mahmud, Shahriar Kanon, S.M. Muntasir Islam, and Kh Shahriya Zaman, *"Environmental Compensation and Physics-Informed Machine Learning Calibration of a Low-Cost Optical Dust Sensor for PM2.5 and AQI Monitoring,"* **Indoor and Built Environment** (Sage Publications). Manuscript ID: IBE-26-0857, submitted September 2026.

Peer-Reviewed Conference Proceedings & Presentations

1. **Samiul Hasan**, Fardous Mahmud, Naim Abrar Shaharia, Hemal Sharma, Minhazul Abedin Nabil, Kh Shahriya Zaman, and Feroz Ahmed, *"An Au/ITO Dual-Layer Photonic Crystal Fiber Surface Plasmon Resonance Biosensor for Label-Free Detection and Discrimination of Basal, HeLa, and Jurkat Cancer Cells,"* 2nd International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS 2026). **(Accepted).**
2. F. Mahmud, **Samiul Hasan**, Naim Abrar Shaharia, Minhazul Abedin Nabil, H. Sharma, and Khosru M. Salim, *"High Step-Up Interleaved Boost Converter with Coupled Inductors and Cascadable Symmetric Voltage Multiplier Cell,"* 2026 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON 2026), Dhaka, Bangladesh, August 2026. **(Certificate of Presentation awarded).**
3. F. Mahmud, **Samiul Hasan**, H. Sharma, S. H. Kanon, E. Haque, and F. Ahmed, *"Numerical Investigation of ZrN-ITO Coated Photonic Crystal Fiber Based Plasmonic Refractive Index Sensor,"* 2025 7th International Conference on Electrical Information and Communication Technology (EICT 2025), Khulna, Bangladesh, December 2025. (10.1109/EICT68394.2025.11355580).

MEMBERSHIPS & VOLUNTARY LEADERSHIP

2025/02/05 – 2025/12/31
Vice Chairperson — IEEE Photonics Society IUB Student Branch

Event & Workshop Leadership: Led the planning and execution of departmental events and technical workshops, including a Radio Astronomy workshop, promoting student engagement, peer learning, and technical knowledge sharing.
Industrial Visit Coordination: Coordinated an educational industrial visit to **BTCL (Bangladesh Telecommunications Company Limited)**, connecting academic learning with practical industry experience and professional exposure.

2024/02/02 – 2024/12/30
Event Coordinator — IEEE IUB Student Branch

1. Coordinated technical workshops, industrial visits, and student engagement activities, developing skills in event planning, teamwork, communication, and logistics.
2. Supported IUB Admission Tests (Spring & Summer 2024) as a volunteer, assisting with examination coordination and candidate management.